Urine dipstick — pads, grades & differentials

all ages · adults, pregnancy and children · casualty / ward / clinic

Urine dipstick · 10 pads

Leucocytes

read 2 min

Neg | Tr | + | ++ | +++

Normal: Negative

- ▸ UTI; leucocytes only → repeat on a 2nd specimen ✚
- ▸ Sterile pyuria: renal TB, interstitial nephritis, calculi, urethritis

FP: vaginal contamination · FN: dilute urine, glycosuria, vit C, nitrofurantoin, rifampicin

Nitrite

read 60 s

Neg | Pos | Pos

Normal: Negative

- ▸ Bacteriuria — usually Gram-negatives (E. coli, Klebsiella, Proteus)
- ▸ Pregnancy: nitrites AND leucocytes → treat empirically ✚

FN: Enterococcus, GC, TB; urine < 4 h in bladder; vit C; pH < 6

Urobilinogen

read 60 s

Norm | ↑ | ↑↑ | ↑↑↑

Normal: Normal trace

- ▸ ↑ in haemolysis (with unconjugated hyperbilirubinaemia, ↑ LDH) ✚
- ▸ ↑ hepatocellular disease; absent in complete biliary obstruction

Neither sensitive nor specific — use LFTs

Protein

read 60 s

Neg | Tr | + | ++ | +++ | ++++

Normal: Negative (Tr 0.15–0.3 · + 0.3–1 · ++++ > 5 g/L)

- ▸ ≥ 1+ → repeat on MSU; if persistent → spot PCR (significant > 0.015 g/mmol) ✚
- ▸ Pregnancy ≥ 20 wk + BP ≥ 140/90: significant = ≥ 2+ on 2 samples ≥ 4 h apart → pre-eclampsia ✚
- ▸ Nephrotic: child ≥ 3+ (PCR ≥ 0.2 g/mmol); adult PCR > 0.25 g/mmol ✚
- ▸ With blood: glomerulonephritis ✚
- ▸ Transient: fever, exercise, UTI, orthostatic

Albumin only — misses light chains (myeloma) · FP: pH > 9, contrast · FN: dilute urine

pH

read 60 s

5 | 6 | 6.5 | 7 | 8 | 8.5

Normal: 5.0–6.0 (range 4.5–8.0)

- ▸ Alkaline: urease organisms (Proteus), stale sample
- ▸ Alkaline: struvite stones, RTA, vomiting · Acid: acidosis/DKA, starvation

Delayed processing ↑ pH (bacteria make ammonia)

Blood

read 60 s

Neg | Tr | + | ++ | +++

Normal: Negative (exclude menses)

- ▸ Schistosomiasis (common), UTI, kidney stone (colic), anticoagulants ✚
- ▸ Painless isolated: bleeding disorder, malignancy, structural, TB (urine cytology, TB culture) ✚
- ▸ Uniform colour, no RBCs on microscopy → haemolysis or rhabdomyolysis (CK)

Blood on the strip ≠ infection ✚

FP: povidone-iodine · FN: vit C · red urine also: beetroot, rifampicin

Specific gravity

read 45 s

1.000 | 1.005 | 1.010 | 1.020 | 1.030

Normal: 1.001–1.035

- ▸ High: dehydration, glucose/protein, contrast · Low: CKD, elderly
- ▸ Low/fixed ~1.010: DI, water load, isosthenuria

Dipstick ↑ if pH < 6, ↓ if pH > 7 — accuracy limited

Ketones

read 40 s

Neg | Tr | + | ++ | +++

Normal: Negative

- ▸ Adult DM with ketonuria > 1+ → urgent same-day referral ✚
- ▸ Child DKA: ketonuria 2+ with glucose > 11 mmol/L ✚
- ▸ Labour dehydration/ketosis; hyperemesis ✚
- ▸ Starvation, alcoholic ketoacidosis, SGLT2i (euglycaemic DKA)

Misses β-hydroxybutyrate — use blood ketones · FN: vit C, NAC

Bilirubin

read 30 s

Neg | + | ++ | +++

Normal: Negative

- ▸ Conjugated bilirubin: hepatitis (before jaundice), cholestasis/obstruction
- ▸ Pure haemolysis → negative (unconjugated is not excreted)

FN: old sample, vit C — accuracy limited

Glucose

read 30 s

Neg | Tr | + | ++ | +++

Normal: Negative — any glucose is abnormal

- ▸ Glycosuria → exclude diabetes mellitus ✚
- ▸ Pregnancy ≥ 1+ on ≥ 2 occasions → GDM risk factor ✚
- ▸ Child DKA: heavy glycosuria ≥ 3+ ✚
- ▸ Normal plasma glucose: SGLT2 inhibitor, renal glycosuria, Fanconi

Appears when plasma glucose > ~10 mmol/L · FN: vit C, ketones, aspirin

PATTERNS

No LEU / NIT → UTI unlikely ✚

LEU or NIT + UTI symptoms → treat; MC&S if possible ✚

LEU + NIT → MC&S and treat empirically ✚

Blood + protein, oedema, BP ↑, brown urine → nephritic: refer ✚

Glucose + ketones → DKA until proven otherwise: HGT, blood ketones, VBG ✚

Bilirubin + with urobilinogen absent → obstructive jaundice

PEARLS & PITFALLS

Child with fever of unknown origin: always test urine (fresh bag specimen) ✚

Bag urine = screening only; positive → aseptic sample (in-out catheter/SPA if ill < 2 y) for MCS ✚

Haematuria > 1+ in hypertension → investigate ✚

Fresh midstream sample; read each pad at its stated time

Pad colours are illustrative — read against the bottle chart at the stated time. Green outline = normal pad.

MSD MSD Manual Prof., Evaluation of the Patient With Renal Issues (urinalysis), updated Feb 2026 · **MSDLIV** MSD Manual Prof., Laboratory Tests of the Liver and Gallbladder, Nov 2025 · **CLASSIC** classic teaching · ✚ **PHC** STG & EML Primary Healthcare 8th ed. 2024 (updated Sept 2026) — p. 8.3, 8.8, 8.10, 8.13, 9.12 · ✚ **HOSP** STG & EML Hospital Level Adults 6th ed. 2024 (updated Mar 2026) — p. 3.38, 6.3, 6.28, 6.32, 7.7, 7.17, 8.14 · ✚ **PAED** STG & EML Paediatric Hospital Level 5th ed. (updated Mar 2026) — p. 3.4, 6.5, 7.19 · ✚ **MAT24** National Integrated Maternal & Perinatal Care Guidelines for SA, 5th ed. 2024 — p. 103
✚ = SA STG / national guideline (pages in footer). Unmarked lines: the other sources listed. Checked against PocketDoc\Guidelines 2026-10-31. **Card** urine-dipstick · updated 2026-10-31 · ANDH district level